DE–8344 **11**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2014.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2002 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What are 'Web Browsers'?

2. List the elements of computers and communication systems.

3. Expand:

   (a) WWW

   (b) EDI

   (c) ISDN

4. What is meant by 'Netiquette'?

5. What is ternary operator in 'C'? Give an example.

6. List any three string built in functions.

7. What are 'structures' in C?

8. How will you intialize arrays? Give examples.

9. Define: Macro.

10. Describe the methods of passing arguments in functions.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the different types of application softwares.

12. Explain the terms communication cannel and data Transmission.

13. Distinguish between 'while' and 'do–while' loop with examples.

14. Write a program to calculate the value of 'nC_r'.

15. Write a program to check whether the given string is palindrome or not. (eg: ADA is a palindrome).

16. Explain 'Inline function' with an example.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Write a note on the following:

    (a) Internet and Intranet

    (b) Modem

    (c) Groupware and work group computing. (5+5+5)

18. Write a C program to find a solution of quadratic equation $ax^2 + bx + c = 0$.

19. Explain the following with examples.

    (a) Pointers

    (b) Operators

    (c) Functions. (5+5+5)

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DE–8345 **12**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2014.

OPERATING SYSTEMS

(2002 Onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Compare workstation with minicomputer and mainframe computer.

2. List down the activities in connection with file management, for which the operating system in responsible.

3. Define:

   (a) Response time

   (b) Turn around time.

4. Distinguish preemptive scheduling from non–preemptive process scheduling.

5. What are four necessary conditions for deadlock prevention?

6. How does maskable interrupt differ from non–maskable interrupt?

7. What is DMA and cycle stealing?

8. Define:
   (a) Rotational latency
   (b) Seek time.

9. What is meant by Belady's anomaly?

10. Write the difference between the first fit strategy and best fit strategy.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Give a note on operating system services.

12. Write about process scheduling concepts.

13. Discuss the critical section problem.

14. Narrate the concept of kernel I/O sub system.

15. Describe the concept of segmentation.

16. With neat diagram, explain the file directory structure.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain any four process scheduling techniques.

18. Give a neat description on bankers algorithm for deadlock avoidance.

19. Narrate the various file allocation methods in detail.

DE–8346 **13**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2014.

OBJECT ORIENTED PROGRAMMING AND C++

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is the objective of Object Oriented Programming?
2. What is encapsulation?
3. What is a scope resolution operator?
4. What is a symbolic constant?
5. Explain the use of constructor.
6. Where can member functions be defined?
7. What is a conditional operator?
8. What do you mean by subscripting operator?
9. What is multiple inheritance?
10. What are virtual functions?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Write a C++ program that will ask for a temperature in Fahrenheit and display it in Celsius.

12. What is a friend function? What are the merits and demerits of using friend functions?

13. Explain nesting of member functions.

14. Explain the use of destructors with an example.

15. Explain how exceptions are handled in C++?

16. Explain the use of "this" pointer with an example.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain the basic concepts in Object Oriented Programming.

18. Explain the various loop structured statements in C++ with examples.

19. Explain overloading of I/O and pointer operators.

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DE–8347 **14**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2014

DATA STRUCTURES AND ALGORITHMS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Define : Data Structure.
2. List the operations performed on Stack.
3. What are strings and its representations?
4. Define : Binary Tree.
5. List the methods of representation of graphs.
6. What is 'Heap'? What is its use?
7. Write a note on 'Hashing'.
8. List the complexity measure of algorithms.
9. Write a note on different 'Searching mechanisms'.
10. What are the advantages and disadvantages of linked lists?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the algorithm of evaluation of a postfix expression using stack with an example.

12. Discuss the various operations associated with Queue.

13. Write 'Pattern matching algorithm'.

14. Explain the different graph traversals.

15. Write and explain selection sort algorithm with an example.

16. Explain 'Search Tree' with an example.

PART C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. (a) Explain the operations on binary trees.

(b) Describe the applications of trees. (9+6)

18. Write and explain the algorithm for polynomial addition with an example.

19. Explain Quick and Radix sort methods.

DE–8348 **15**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2014.

OBJECT ORIENTED DBMS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL the questions.

1. What is meant by data independence?
2. Specify the role of database management systems.
3. What is meant by weak entities?
4. What is aggregation?
5. Write notes on ODBC.
6. Write notes on JDBC.
7. Write note on database tuning.
8. Define the term functional dependency.
9. What is meant by mandatory access control?
10. Write notes on XML DTDS.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the levels of abstraction in DBMS in detail.
12. Describe the complex integrity constraint in SQL in detail.

13. Discuss about conceptual database design with the ER model.

14. Write in detail about fourth normal form with examples.

15. Describe the architecture of parallel databases in detail.

16. Discuss about the database design for an ORDBMS.

PART C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Discuss about parallel query optimization and distributed query processing.

18. Describe the transaction management in DBMS in detail.

19. Discuss about nested queries and query by example (QBE) with examples.

DE–8349 **21**

DISTANCE EDUCATION

M.Sc.(Information Technology) DEGREE EXAMINATION,
MAY 2014.

INTERNET PROGRAMMING AND WEB DESIGN

(2002 Onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL the questions.

1. Define the term : Internet.

2. What are web browsers?

3. What is JDK?

4. List the features of Perl.

5. List the Arithmetic operators in Java script.

6. Define : Sub procedure in VB script.

7. Write a note on : Script tag in HTML.

8. Define : ActiveX control.

9. Write the use of <ul> tag in HTML.

10. Define : Animation.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Write about WWW design issues.

12. Explain the data types in java.

13. Write short notes on : CGI.

14. Explain about functions in java script with examples.

15. Write short notes on : SGML.

16. Explain about dynamic graphics.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain about developing intranet applications.

18. Write a Java script to find the number of occurrences of a given number in a set of stored numbers.

19. Explain creating Netscape Navigator plug-ins.

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DE–8350 **22**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2014.

COMPUTER NETWORKS

(2002 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Distinguish multicasting from broadcasting.

2. List down seven layers of OSI reference model.

3. What is meant by Quadrature amplitude modulation?

4. Expand: (a) DTE (b) DCE. Also state the difference between them.

5. What is bit stuffing? Illustrate.

6. Define the term, "Petri net" with its elements.

7. Write the difference between adaptive and non-adaptive algorithms.

8. State the importance of TCP and UDP protocols.

9. Mention the roles of user agent and message transfer agent.

10. Differentiate monoalphabetic substitution from polyalphabetic cipher.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Give a note on network architectures with neat diagram.

12. With neat diagram, explain the working principle of ISDN system architecture.

13. Describe the concepts of carrier sense multiple access protocols and collision free protocols.

14. Write notes on the following connection management procedure :

    (a) Flow control and buffering (6)

    (b) Multiplexing. (4)

15. Discuss the concept of Internetworking in detail.

16. Explain the importance of multimedia with JPEG and MPEG standards.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Write notes on :

    (a) Networks structure (7)

    (b) Uses of computer networks. (8)

18. List and explain the various data link layer design issues in detail.

19. Explain the following routing algorithms :

    (a) Shortest path routing

    (b) Hierarchical routing

    (c) Broadcast routing.

DE–8351 **23**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2014.

SOFTWARE ENGINEERING

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Draw the common process framework of a software.

2. Who proposed the spiral model? Also state six task regions of spiral model.

3. What are the environmental resources?

4. Write the process of identifying risks.

5. What is meant by degree of rigor? State its types.

6. Distinguish internal failure costs from external failure costs.

7. Define : Stepwise refinement. Who proposed it?

8. Differentiate Data Flow Diagram from E-R Diagram.

9. Give definitions for
   (a) Smoke testing
   (b) Top down integration testing.

10. Define :
    (a) Stress testing
    (b) Performance testing.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Discuss the software process model, "Spiral Model" in detail.

12. With an example, explain the problem-based software product estimation.

13. Write about software reviews in detail.

14. Give a note on software prototyping.

15. With neat diagram, explain the process of cyclomatic complexity.

16. Explain the need for validation and debugging.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Give a neat description on various risk estimation of a software project.

18. Explain the concept of formal technical reviews in detail.

19. Describe the process of functional modeling and information flow.

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DE–8352 **24**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2014.

VISUAL PROGRAMMING

(2002 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Name any three components of a window.

2. What is meant by SDI?

3. List three properties of a form.

4. Bring out the differences between a function and a procedure.

5. What is the role of a check box?

6. What is the need for a dialog box?

7. What is an Object?

8. What is a splitter window?

9. Expand :

   (a) DLL

   (b) DDE.

10. Specify the importance of ODBC.

SECTION B — ($4 \times 10 = 40$ marks)

Answer any FOUR questions.

11. What is a device context? Explain the objects and their properties.

12. Explain the development of a form with its properties and methods.

13. Explain the looping structures of VB with examples.

14. How will you create a menu in VB? Explain.

15. Explain the Document view architecture.

16. What is OLE? Explain.

SECTION C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Explain the steps in the creation of a simulated calculator with VB.

18. Discuss the controls available in a Tool box.

19. Discuss the exception handling mechanism in detail.

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DE–8353 **25**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2014.

MULTIMEDIA AND APPLICATIONS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Write short notes on multimedia.

2. What is a system component in multimedia? Explain.

3. Define and discuss about text.

4. State the distinct features of digital ink.

5. Write short notes on JPEG.

6. Discuss about Digital video.

7. Write short notes on Media classes.

8. What is format class? Shortly discuss about it.

9. What are the media types? Discuss about it.

10. Discuss about Media organization.

PART B — ($4 \times 10 = 40$ marks)

Answer any FOUR questions.

11. Discuss cross platform compatibility.

12. Write notes on speech audio.

13. What is redundancy and visibility? Explain.

14. Write notes on transform classes.

15. Explain CD-family.

16. Write short notes on multimedia standards.

PART C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. What are the commercial tools? Explain

18. Explain the non temporal media types.

19. Explain the various types of compression techniques.

DE–8811 **16**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2014.

PRACTICAL LAB – I – C++

(2002 onwards)

Time : Three hours Maximum : 100 marks

Break-up of marks :

Record	: 10
Algorithm	: 10
Program	: 50
Debugging	: 10
Execution	: 10
Result	: 10
Total	: 100

Examiner has to choose any ONE question.

1. Write a program in C++ to find the total and difference of length of given two tubes specified in meters and centimeters. Use operator overloading.

---------- Cut here ----------

2. Write a C++ program (menu driven) to add and subtract given two matrices of order m×n defined in class, using operator overloading.

---------- Cut here ----------

3. Create a class and write a C++ program to get n names and display them in sorted order. Use pointers.

4. Write a program in C++ to reverse a string and find whether the given string is a palindrome.

---------- Cut here ----------

5. Write a C++ program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

---------- Cut here ----------

6. Write a C++ program to implement the concept of constructor and destructor.

---------- Cut here ----------

7. Write a program in C++ to find whether a given number is an Armstrong number or not.

---------- Cut here ----------

8. Write a program in C++ to find whether a number is prime or not.

---------- Cut here ----------

9. Write a program in C++ to calculate the salary of employees. Define your own problem and suitable classes.

---------- Cut here ----------

10. Write a C++ program to calculate the electricity bill by defining the problem and suitable classes.

DE–8812 **17**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2014.

First Year

LAB II — DATA STRUCTURES LABORATORY

(2002 onwards)

Time : Three hours Maximum : 100 marks

ONE question should be selected by the Examiner and given to the candidates by lot system.

Each question carries 50 marks.

Break-Up of Marks

Record	:	10
Algorithm	:	10
Program	:	50
Debugging	:	10
Execution	:	10
Result	:	10
Total	:	100

1. (a) Write a C++ program to add and subtract the two given $m \times n$ matrix.

   (b) Write a C++ program for implementing stack operations using array.

2. (a) Write a C++ program to add and multiply the two given $m \times n$ matrix.

   (b) Write a C++ program for creation of linked list and performing its operation using array.

---------- Cut here ----------

3. (a) Write a C++ program to sort the following list using Bubble sort. –16, 20, –46, 0, 34, –38, –25, 20, 17.

   (b) Write a C++ program to copy the content of one file to another file.

---------- Cut here ----------

4. (a) Write a C++ program to search an element using Binary search.

   (b) Write a C++ program for creation of linked list and performing its operations using pointers.

---------- Cut here ----------

5. (a) Write a C++ program to sort elements using insertion sort.

   (b) Write a C++ program for implementing Queue structure and its operations using Array.

DE–8813 **26**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2014.

INTERNET PROGRAMMING LAB

(2002 onwards)

Time : Three hours Maximum : 100 marks

Break-Up of Marks

Record	:	10
Program	:	50
Form Design	:	10
Debugging	:	10
Execution	:	10
Result	:	10
Total	:	100

Examiner to select and give one question to each candidate.

1. (a) Write a Java script program which will implement mouse over event by opening a new document when a mouse pointer moves over a link and test your program.

   (b) Using database connectivity display records in a table.

---------- Cut here ----------

2. (a) Write a java program using applet to display a line using different colors.

   (b) Write a Swing program to create buttons with

      (i) Image

      (ii) Tool tip.

3. (a) Write a VB script to do the following :

   (i) Calculate the leap year

   (ii) Check whether the string is palindrome or not.

   (b) Write a swing program to create images and buttons.

---------- Cut here ----------

4. (a) Write a java script program to create a calculator.

   (b) Write a java program to implement shapes using applets.

---------- Cut here ----------

5. (a) Create a data base for income tax processing.

   (b) Create a HTML document to create a table using table tags.

---------- Cut here ----------

6. (a) Write a swing program to create label, text box and images.

   (b) Create a java script program to verify email address.

---------- Cut here ----------

7. (a) Create a java script program to verify email address.

   (b) Write a java script program to read the student marks for five subjects and print the total and average.

8. (a) Write a VB script program to scroll the text on status bar.

   (b) Write a java program to generate random numbers using random class and arrange the numbers in ascending order and descending order.

---------- Cut here ----------

9. (a) Write a java script that implements string concatenation.

   (b) Design a webpage using HTML code to display the details of air reservation form.

DE–8814 **27**

DISTANCE EDUCATION

M.Sc. (Information Technology)DEGREE EXAMINATION, MAY 2014.

LAB IV — VISUAL PROGRAMMING LAB

(2002 onwards)

Time : Three hours Maximum : 100 marks

Examiner to select and give ONE question to each candidate.

1. (a) In VB, Write event procedure for the following

   (i) To display date and time in a text box

   (ii) To find the power of a number

   (b) Write a VC++ program to create a window of desired size using MFC.

---------- Cut here ----------

2. (a) In VB, Write event procedure to perform the following.

   (i) Reverse a string

   (ii) To determine if a given string is palindrome or not

   (b) Write a VC++ program to handle windows messages in MFC program.

3. (a) Using text box, combo box and command buttons in VB, write a program to add a list of country names during execution. Include provisions for deleting, searching and sorting the names.

(b) Write a VC++ program to fill background of the client area with a bitmap.

---------- Cut here ----------

4. (a) Write a VB program, to create a simple calculator.

(b) Write a VC++ program , to get the status of the shift and toggle keys using MFC.

---------- Cut here ----------

5. (a) In VB, Using Flex grid control, display the addition table of 20 rows and colums.

(b) Write a VC++ program to generate a status bar and show the status of the caps lock, Num lock, and Scroll lock keys in it.

---------- Cut here ----------

6. (a) In VB, create a notepad with file and edit menu operations.

(b) Write a VC++ program to create a list box in a window.

---------- Cut here ----------

7. (a) Create a table with employee details for preparing salary slips, with provisions for adding, deleting searching and modifying records.

(b) Write a VC++ program to find out if a mouse is present and if so, how many buttons it has.

8. (a) Create a table with patient details for maintaining patient information, with provisions for adding, deleting, searching and modifying records.

(b) Write a VC++ program to find out if a mouse is present and if so, how many bottons it has.

---------- Cut here ----------

9. (a) Write a VB program to insert, delete and search records in a random access file.

(b) Write a VC++ program to generate a status bar and show the status of the Caps lock. Num lock, and Scroll lock keys in it.

---------- Cut here ----------

10. (a) Create a VB application with a drivelistbox, dir listbox to do the following.

(i) Select an exe file and execute it

(ii) Select the picture and load it to the form

(iii) Filter the files in the file list box, according to the extension chosen in the combo box.

(b) Write a VC++ program, to get the status of the shift and toggle keys using MFC.